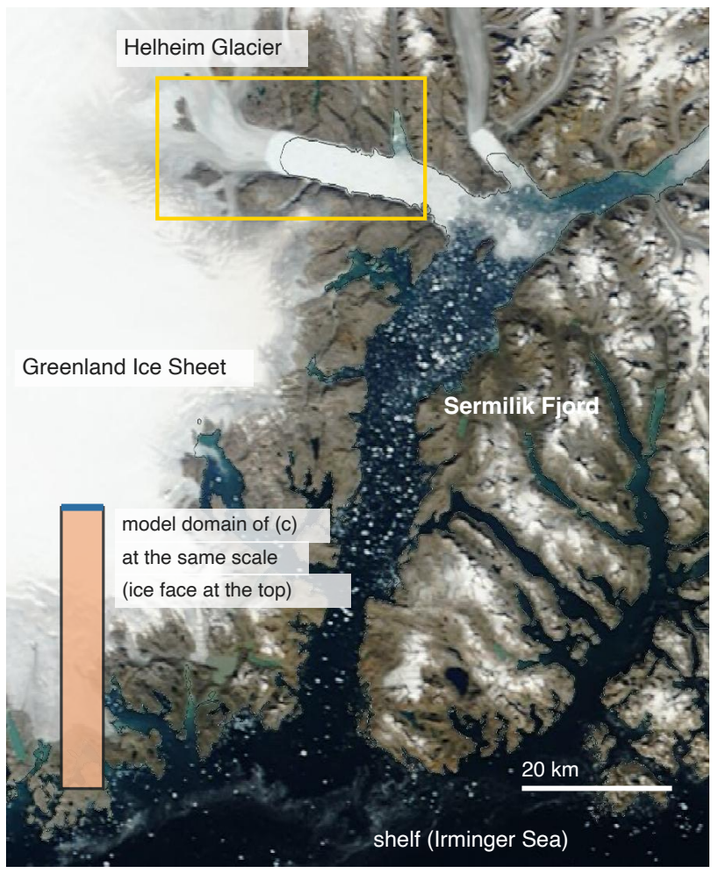
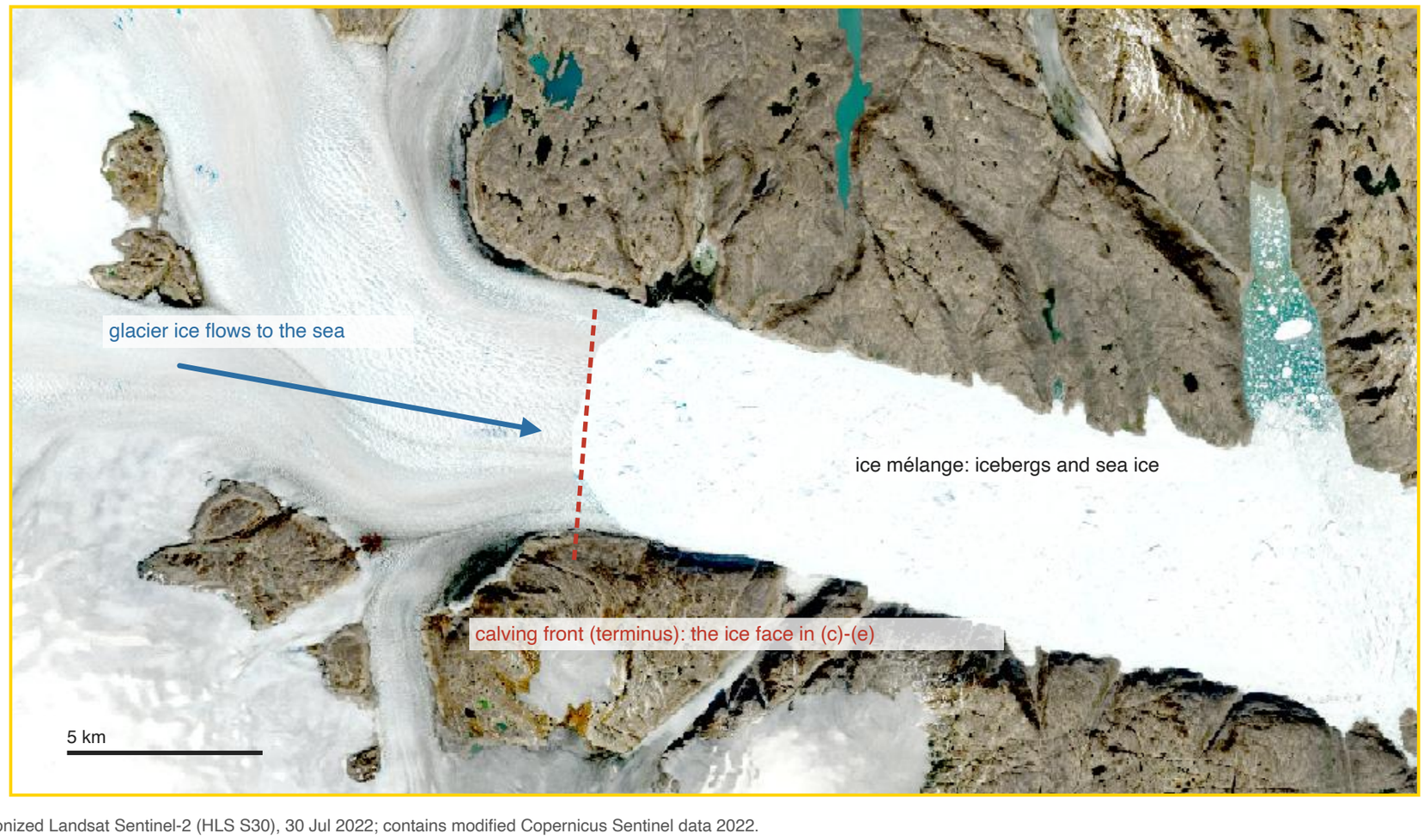
(a) Sermilik Fjord and Helheim Glacier, SE Greenland

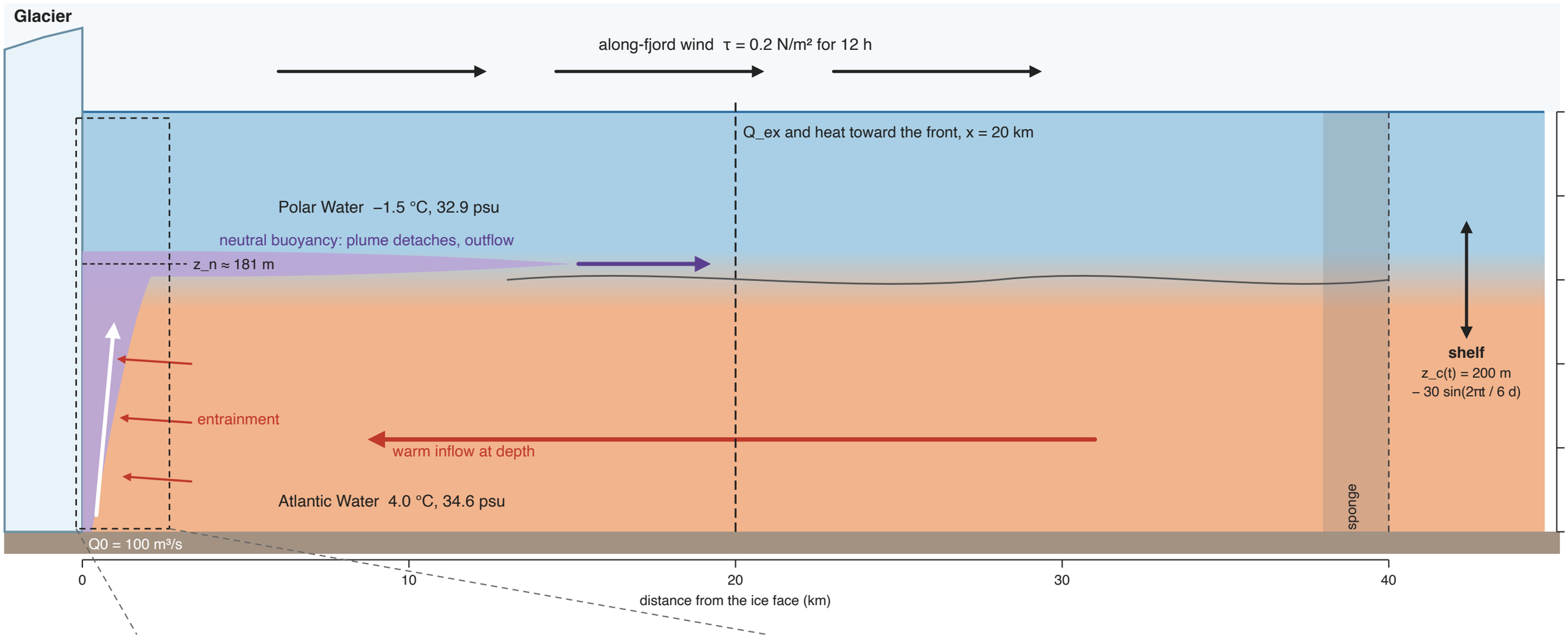


(b) Helheim Glacier terminus, 30 m resolution

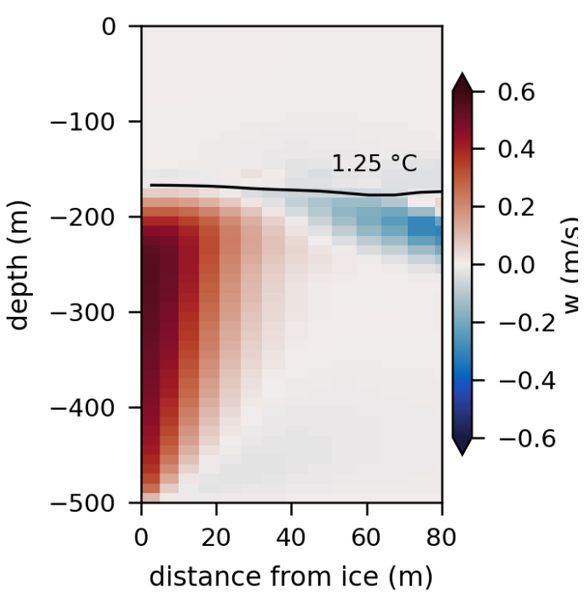


Imagery: NASA Worldview / GIBS. (a) MODIS Terra true colour, 5 Aug 2022. (b) Harmonized Landsat Sentinel-2 (HLS S30), 30 Jul 2022; contains modified Copernicus Sentinel data 2022.

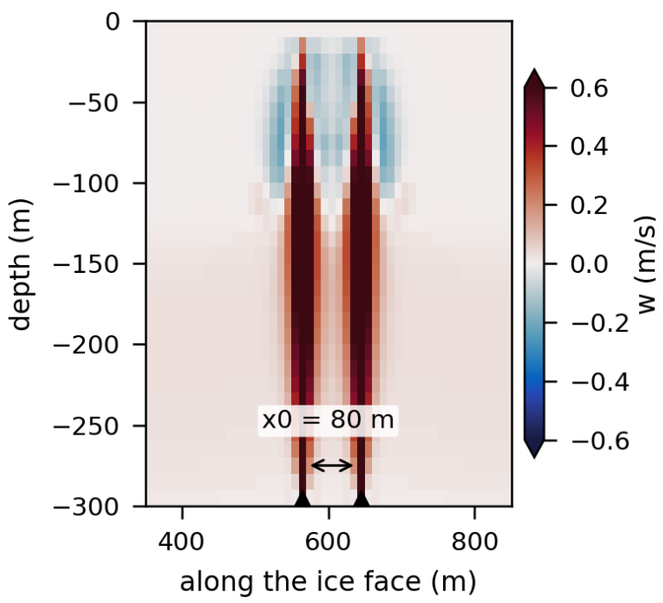
(c) Along-fjord section of the idealised fjord, configuration (c); vertical scale exaggerated, plume enlarged



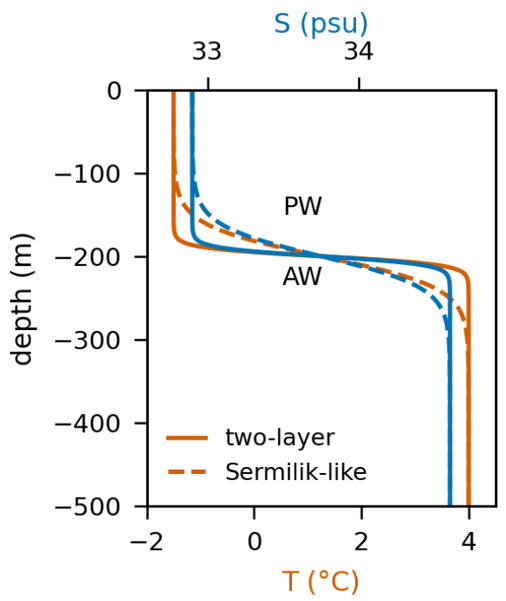
(d) config (a): resolved line plume



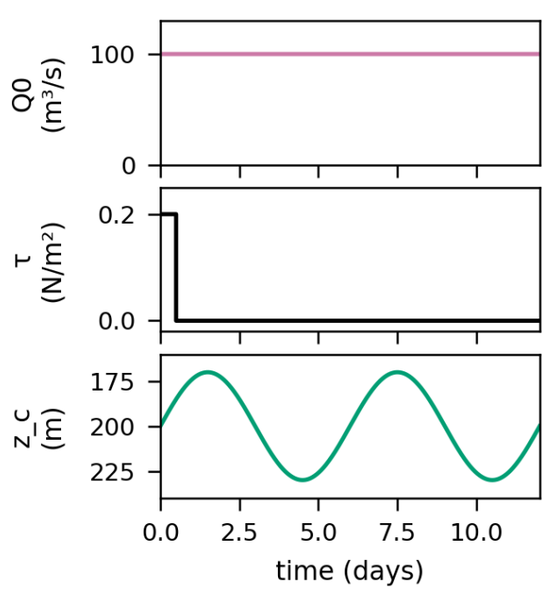
(e) config (b): two plumes on the ice face



(f) initial T and S profiles



(g) forcing of config (c)



(d), (e): model output of this experiment (w, vertical velocity; black line in (d), the 1.25 °C isotherm between PW and AW). Schematic in (c) not to scale.